
one step, no host round trip: captured as one CUDA graph by lodos.step.make_step_graph
lodos.rbd
kinematics, $M = LL^\top$,
$\ddot{q}_s = M^{-1}(\tau - b)$
collision
dcol: Lodos's
d.contact: MJWarp's
export
Lodos's contacts into d.contact:
each, reduced, or as groups
rows
M gathered into d.M,
make_constraint
MJWarp's Newton solve
its own, fused, or in the rows;
exact mode: in a proximal loop
$\ddot{q}$, written
in place
semi-implicit Euler
$\dot{q}^+ = \dot{q} + \Delta t\, \ddot{q}$,
$q^+ = q \oplus \Delta t\, \dot{q}^+$
or implicitfast
contact dynamics: SolverDriver.contact_dynamics (lodos.cd.driver), the NLSP solver